PRACTICAL-3

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3.1.1. Numpy Array Operations

Write a Python program to create a NumPy array based on user input and display the following:

- The created NumPy array.
- The number of dimensions of the array using ndim
- The shape of the array using shape
- The total number of elements in the array using size

  Assume all input elements are valid integers.

Input Format:

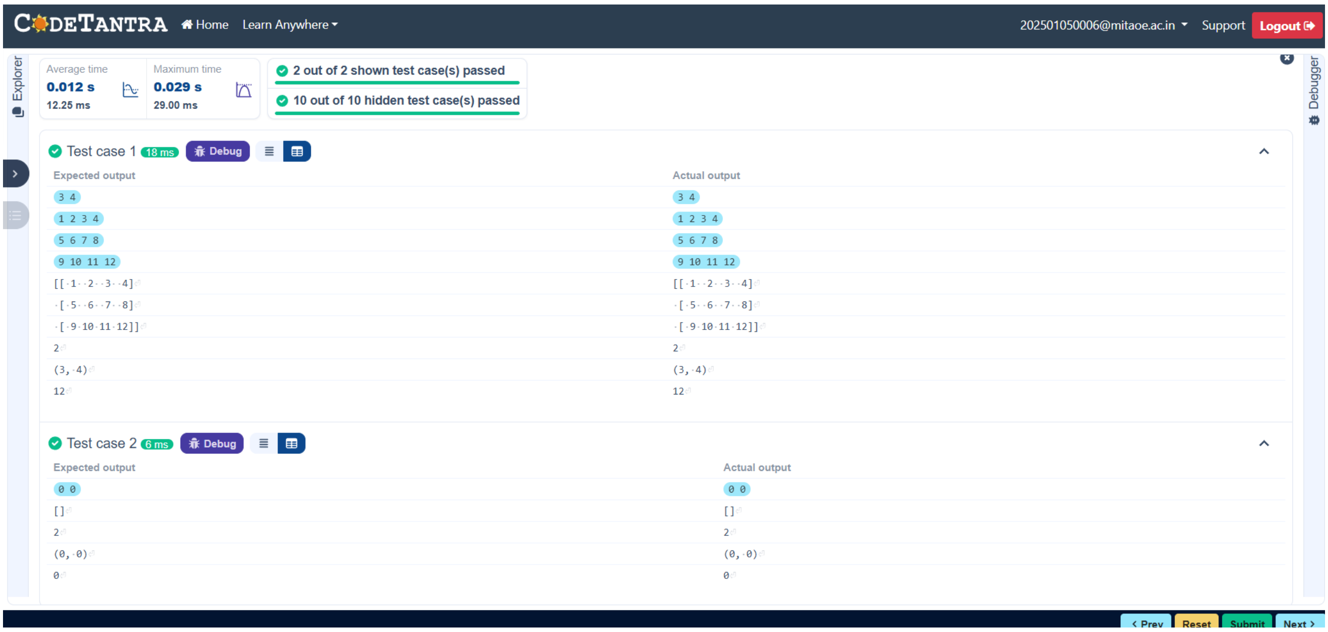
- The first line contains two space-separated integers, and , representing the number of rows and the number of columns.
- The next lines contain space-separated integers representing the elements of the array, entered row by row.

Output Format:

- The created NumPy array (printed as a 2D array).
- An integer representing the number of dimensions of the array.
- A tuple representing the shape of the array.
- An integer representing the total number of elements in the array.

CODE:

```
import numpy as np rows, cols =
map(int, input().split()) elements =
[] for _ in range(rows) :
        elements.extend(map(int, input().split()))
array = np.array(elements).reshape(rows, cols)
print(array) print(array.ndim) print(array.shape)
print(array.size)
```



<u>3.2.1. Numpy: Matrix Operations</u>

The given code takes two matrices, , and , as input from the user and converts them into NumPy arrays.

Task:

You are required to compute and display the results of the following matrix operations:

1. **Addition** ()
2. **Subtraction** ()

3. **Element-wise Multiplication** ()
4. **Matrix Multiplication** ()
5. **Transpose of Matrix A Input Format:**

- The user will input 3 rows for , each containing 3 integers separated by spaces.
- Similarly, the user will input 3 rows for , each containing 3 integers separated by spaces.

Output Format:

The program should display the results of the operations in the following order:

1. The result of Addition.
2. The result of Subtraction.
3. The result of Element-wise Multiplication.
4. The result of Matrix Multiplication.
5. The Transpose of Matrix A.

CODE

import numpy as np # Input matrices print("Enter Matrix A:")

matrix_a = np.array([list(map(int, input().split())) for i in

range(3)]) print("Enter Matrix B:") matrix_b =

np.array([list(map(int, input().split())) for i in range(3)])

Addition addition_result = matrix_a + matrix_b

print("Addition (A + B):") print(addition_result) #

Subtraction subtraction_result = matrix_a - matrix_b

print("Subtraction (A - B):") print(subtraction_result) #

Multiplication (element-wise)

elementwise_multiplication_result = matrix_a * matrix_b

print("Element-wise Multiplication (A * B):")

print(elementwise_multiplication_result) # Matrix

multiplication (dot product)

matrix_multiplication_result=np.dot(matrix_a, matrix_b)

print("A dot B:")

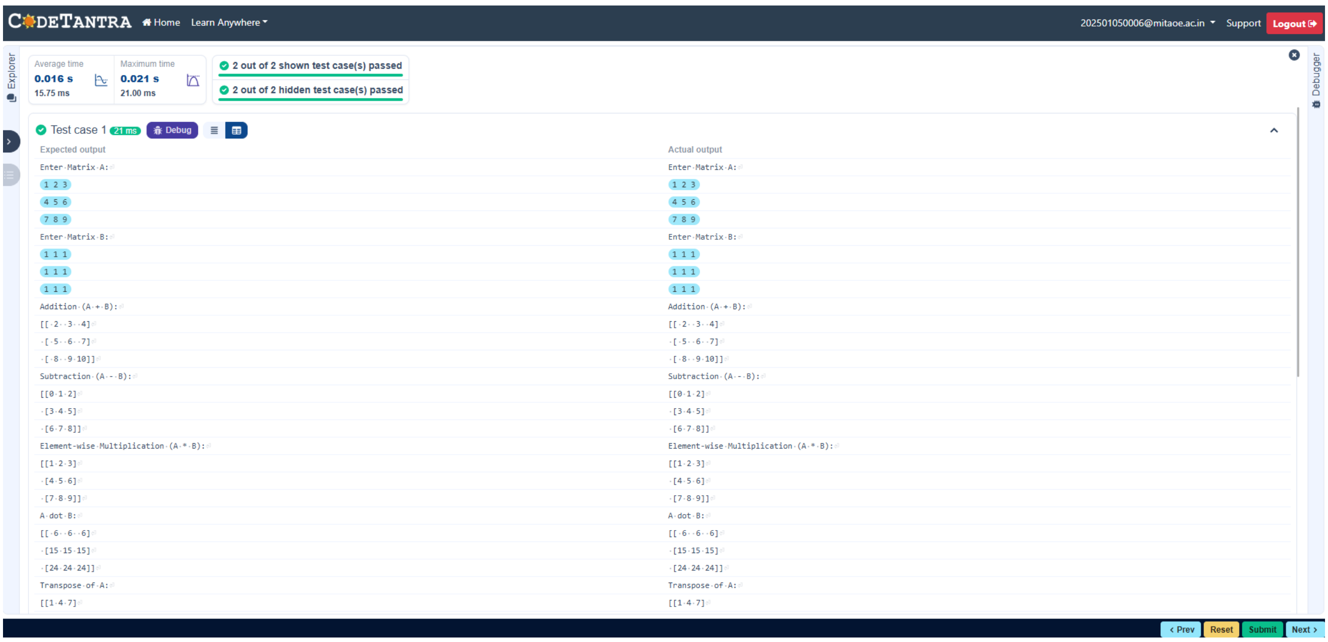
print(matrix_multiplication_result)

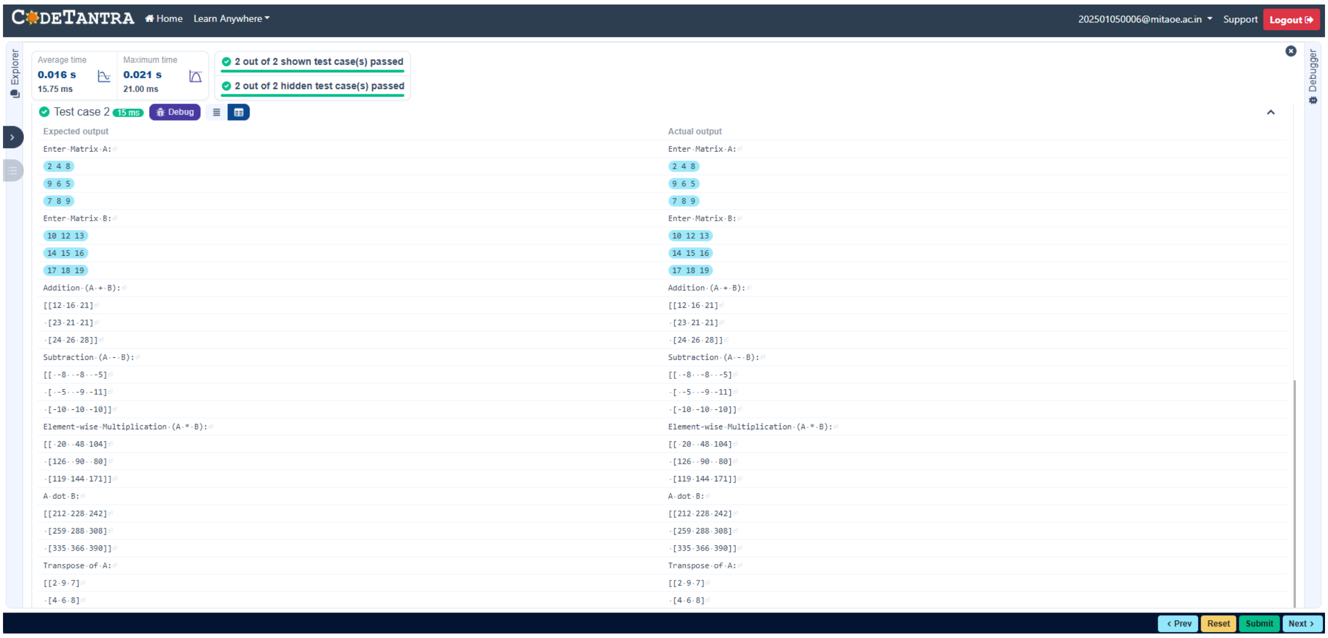
Transpose transpose_a

= matrix_a.T

print("Transpose of A:")

print(transpose_a)





3.2.2. Numpy: Horizontal and Vertical Stacking of Arrays

You are given two arrays arr1 and arr2. You need to perform horizontal and vertical stacking operations on them using NumPy.

- **Horizontal Stacking**: Stack the two matrices horizontally (side by side).
- **Vertical Stacking**: Stack the two matrices vertically (one below the other).

Input Format:

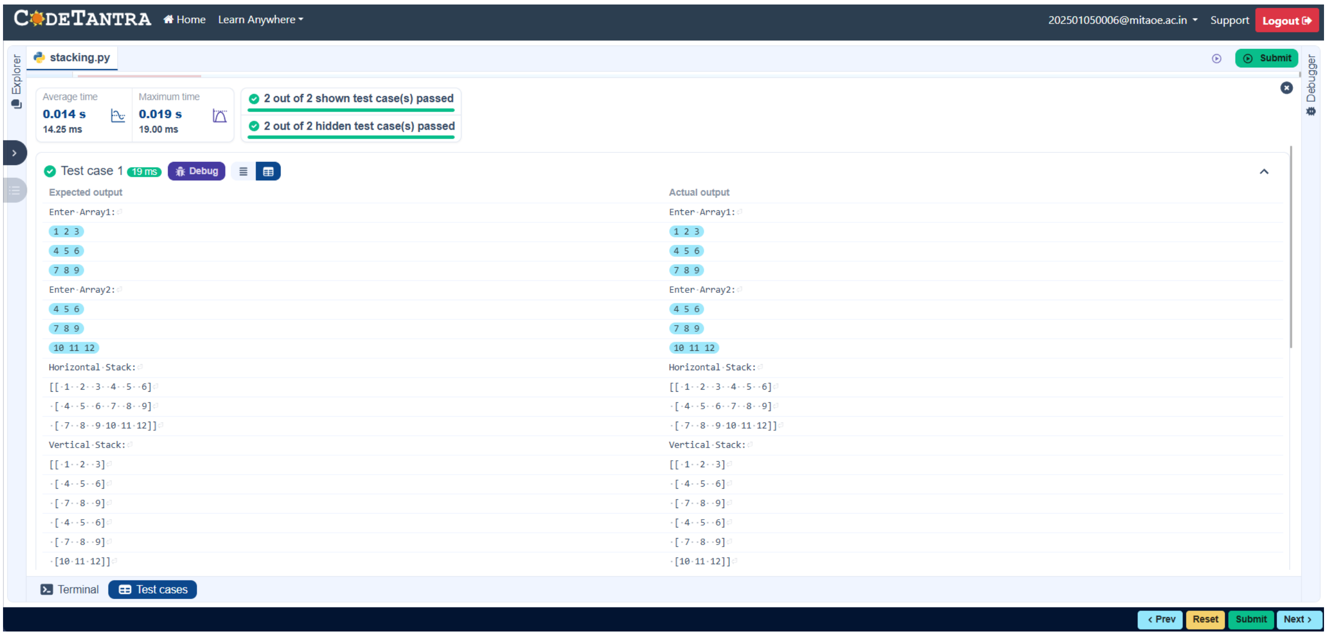
- The program should first prompt the user to input two 3x3 arrays.
- Each array consists of 3 rows, and each row contains 3 space-separated integers.
- The user will input the two arrays row by row.

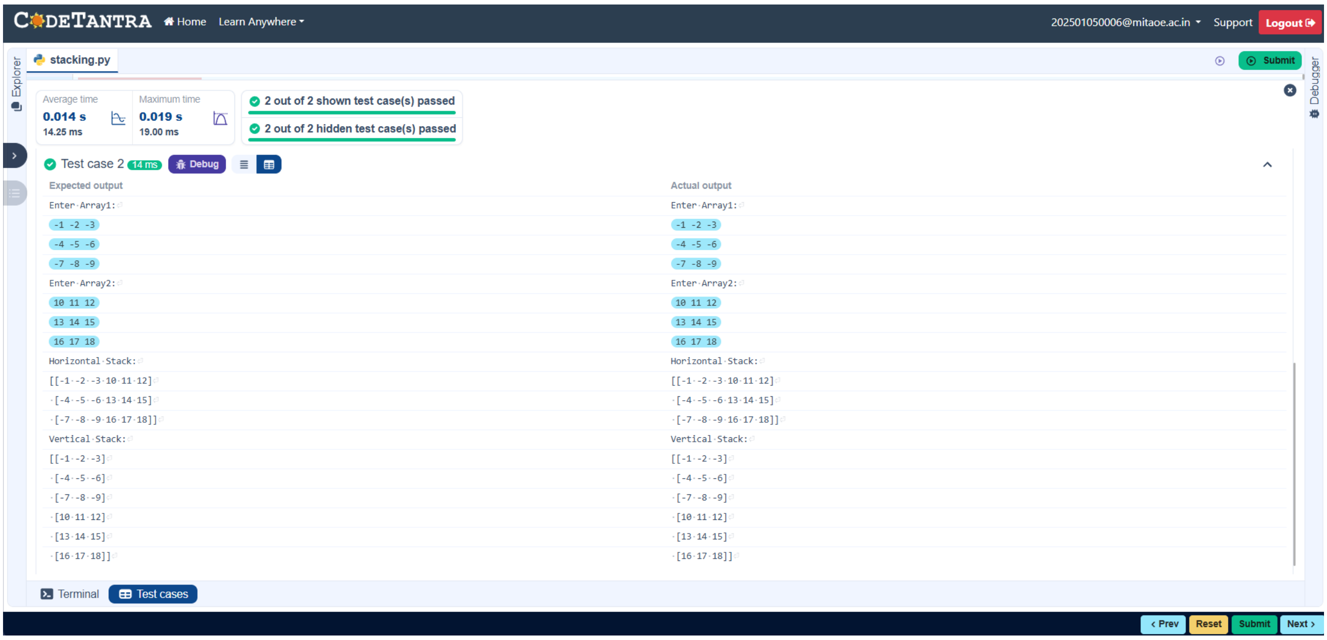
Output Format:

- The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.
- The program should then display the result of the Vertical Stack (one below the other) of the two arrays.

CODE:

```
import numpy as np # Input matrices print("Enter
Array1:") arr1 = np.array([list(map(int, input().split())) for i
in range(3)]) print("Enter Array2:") arr2 =
np.array([list(map(int, input().split())) for i in range(3)])
# Perform horizontal stacking
(hstack) a=np.hstack((arr1,arr2))
print("Horizontal Stack:") print(a)
# Perform vertical stacking (vstack)
b=np.vstack((arr1,arr2)) print("Vertical
Stack:")
```





3.2.3. Numpy: Custom Sequence Generation

Write a Python program that takes the following inputs from the user:

- Start value: The starting point of the sequence.
- Stop value: The sequence should end before this value.

- Step value: The increment between each number in the sequence.

The program should then generate a sequence using numpy based on these inputs and print the generated sequence. Input Format:

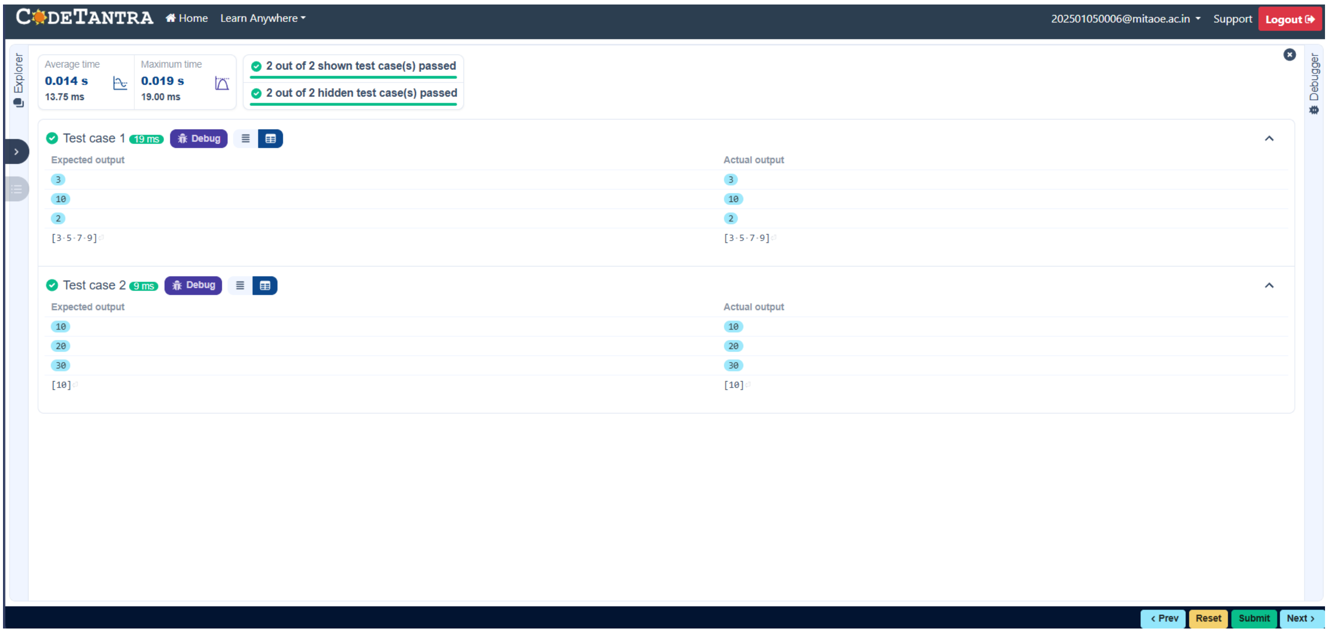
- The user will input three integer values: start, stop, and step, each on a new line.

Output Format:

- The program should print the generated sequence based on the input values.

CODE:

```
import numpy as np

# Take user input for the start, stop, and step of the

sequence start = int(input()) stop = int(input()) step =

int(input())

# Generate the sequence using np.arange()

a= np.arange(start,stop, step) print(a)

# Print the generated sequence
```



3.2.4. Numpy: Arithmetic and Statistical Operations, Mathematical Operations, Bitwise Operators

You are given two arrays A and B. Your task is to complete the function array_operations, which will convert these lists into NumPy arrays and perform the following operations:

1. Arithmetic Operations:

- Compute the element-wise sum, difference, and product of the two arrays.

2. Statistical Operations:

- Calculate the mean, median, and standard deviation of array A.

3. Bitwise Operations:

- Perform bitwise AND, bitwise OR, and bitwise XOR on the arrays (ex: A_i OR B_i).

Input Format:

- The first line contains space-separated integers representing the elements of array A.
- The second line contains space-separated integers representing the elements of array B.

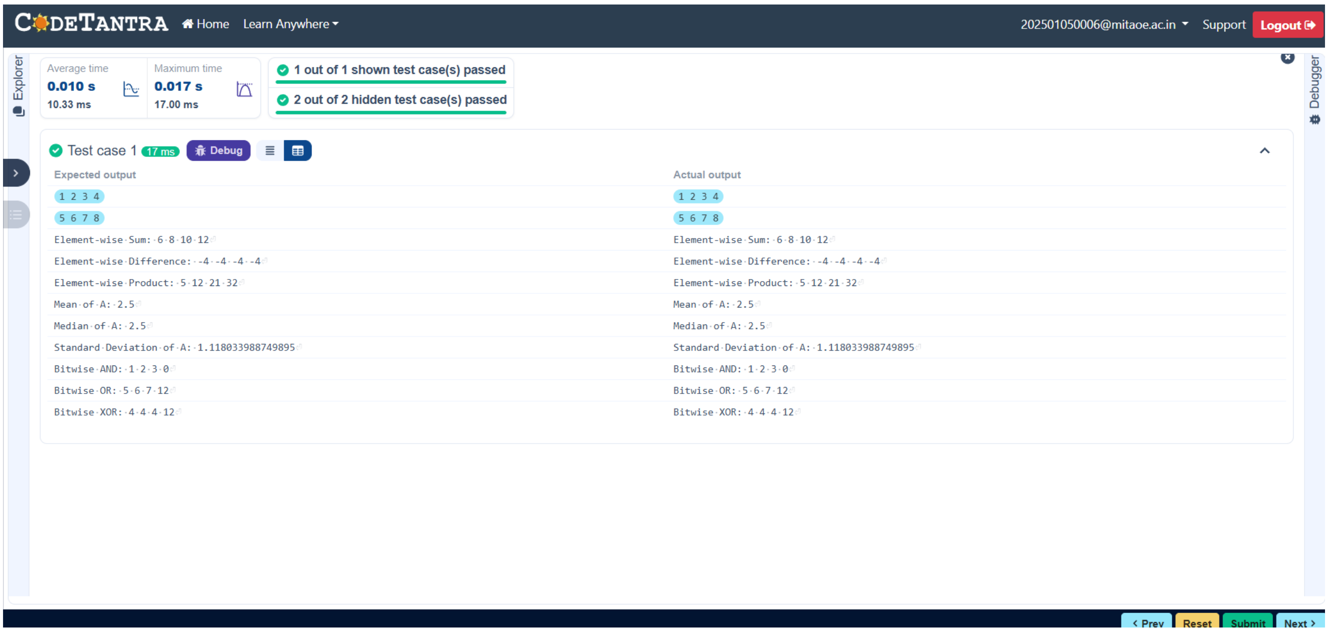
Output Format:

- For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.

CODE:

```
import numpy as np def
array_operations(A, B):
            # Convert A and B to NumPy arrays
        A= np.array(A)
        B= np.array(B)    # Arithmetic
          Operations        sum_result = A + B
           diff_result = A - B        prod_result
          = A * B
           # Statistical Operations
     mean_A = np.mean(A)
median_A = np.median(A)
std_dev_A = np.std(A)
            # Bitwise Operations
    and_result = A & B
    or_result = A | B
     xor_result
```

```
            # Output results with one space between each element
print("Element-wise Sum:", ' '.join(map(str, sum_result)))
print("Element-wise Difference:", ' '.join(map(str, diff_result)))
print("Element-wise Product:", ' '.join(map(str, prod_result)))
print(f"Mean of A: {mean_A}")
    print(f"Median of A: {median_A}")
print(f"Standard Deviation of A: {std_dev_A}")
            print("Bitwise AND:", ' '.join(map(str,
and_result)))    print("Bitwise OR:", ' '.join(map(str,
or_result)))     print("Bitwise XOR:", ' '.join(map(str,
xor_result))) A = list(map(int, input().split()))  #
Elements of array A B = list(map(int, input().split()))
# Elements of array B
array_operations(A, B)
```



3.2.5. Numpy: Copying and Viewing Arrays

The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

- Creating a view of the original_array and assigning it to view_array.
- Creating a copy of the original_array and assigning it to copy_array.

After completing these steps, observe how modifying the view affects the original_array, while modifying the copy does not.

Input Format:

- A single line of space-separated integers.

Output Format:

- After modifying the view:

Original array after modifying view: <**original_array**>

View array: <**view_array**>

- After modifying the copy:

Original array after modifying copy: <**original_array**>

Copy array: <**copy_array**>

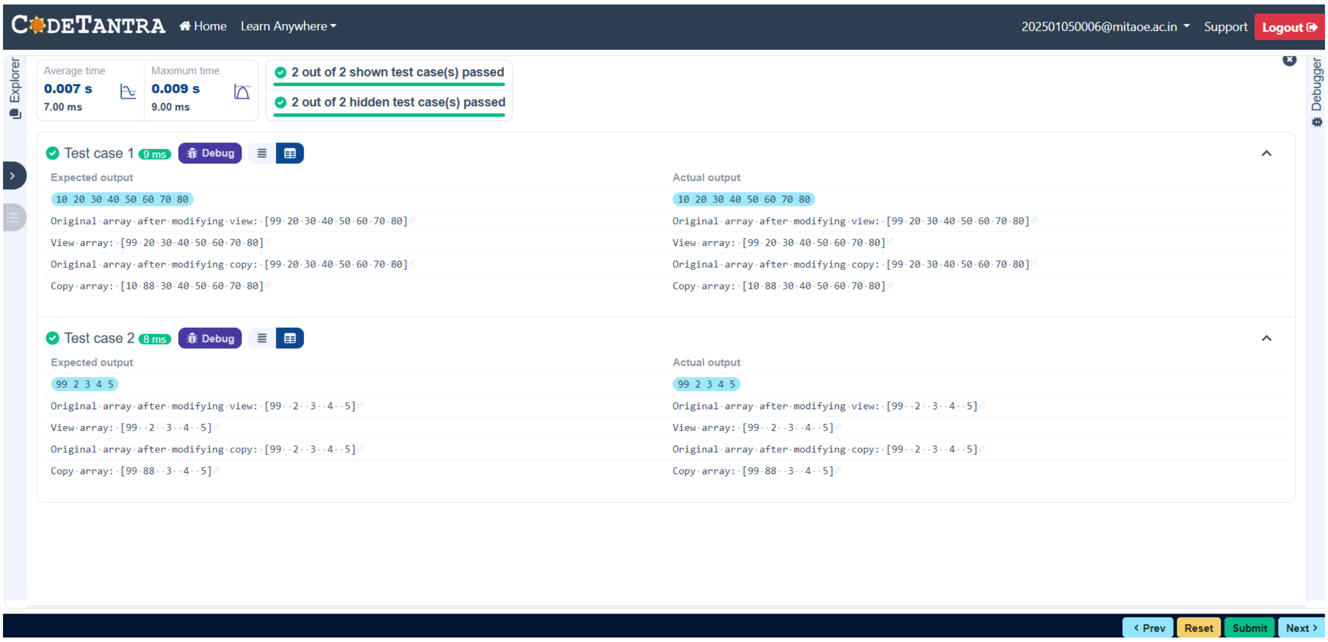
CODE:

```
import numpy as np
inputlist = list(map(int,input().split(" "))
)# Original array original_array
= np.array(inputlist) # Create a
view view_array
=original_array.view()
```

Create a copy copy_array =original_array.copy()

Modify the view view_array[0] = 99 print("Original array after modifying view:", original_array) print("View array:", view_array)

Modify the copy copy_array[1] = 88 print("Original array after modifying copy:", original_array) print("Copy array:", copy_array)



3.2.6. Numpy: Searching, Sorting, Counting, Broadcasting

The given code in the editor takes a single array, array1, as spaceseparated integers as input from the user.

Additionally, it takes the following inputs:

- search_value: The value to search for in the array.

- count_value: The value to count its occurrences in the array.
- broadcast_value: The value to add for broadcasting across the array.You need to complete the code to perform the following operations:

1. **Searching**: Find the indices where search_value appears in array1 and print these indices.
2. **Counting**: Count how many times count_value appears in array1 and print the count.
3. **Broadcasting**: Add broadcast_value to each element of array1 using broadcasting, and print the resulting array.
4. **Sorting**: Sort array1 in ascending order and print the sorted array.

Input Format:

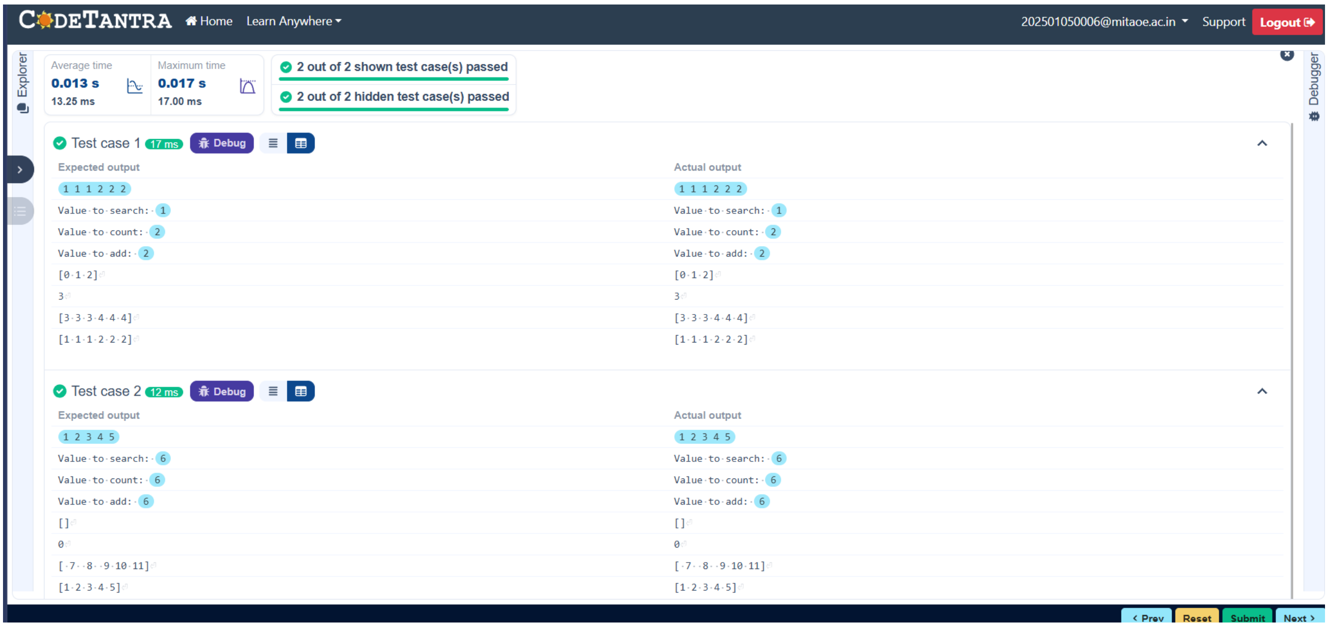
1. A single line containing space-separated integers representing array1.
2. An integer search_value represents the value to search for in the array.
3. An integer count_value represents the value to count in the array.
4. An integer broadcast_value represents the value to add to each element of the array.

Output Format:

1. The indices where search_value occurs in array1.
2. The count of occurrences of count_value in array1.
3. The array after adding the broadcast_value to each element.
4. The sorted array.

CODE:

```
import numpy as np# Input array from the user
array1 = np.array(list(map(int, input().split())))
# Searching search_value = int(input("Value
to search: ")) count_value =
int(input("Value to count: "))
broadcast_value = int(input("Value to add:
")) # Find indices where value matches in
array1 a=np.where(array1==search_value)[0]
print(a)
# Count occurrences in array1
b=np.count_nonzero(array1==count_value) print(b)
# Broadcasting addition c=array1
+ broadcast_value print(c)
# Sort the first array
d=np.sort(array1) print(d)
```



3.2.7. Student Data Analysis and Operations

Write a Python program that takes the file name of a CSV file containing student details, including roll numbers and their marks in three subjects as input, reads the data, and performs the following operations:

- **Print all student details**: Display the complete details of all students, including roll numbers and marks for all subjects.
- **Find total students**: Determine the total number of students in the dataset.
- **Print all student roll numbers**: Extract and print the roll numbers of all students.
- **Print Subject 1 marks**: Extract and print the marks of all students in Subject 1.
- **Find minimum marks in Subject 2**: Identify the lowest marks in Subject 2.
- **Find maximum marks in Subject 3**: Identify the highest marks in Subject 3.
- **Print all subject marks**: Display the marks of all students for each subject.
- **Find total marks of students**: Compute the total marks for each student across all subjects.
- **Find the average marks of each student**: Compute the average marks for each student.
- **Find average marks of each subject**: Compute the average marks for all students in each subject.
- **Find average marks of Subject 1 and Subject 2**: Compute the average marks for Subject 1 and Subject 2.

- **Find average marks of Subject 1 and Subject 3**: Compute the average marks for Subject 1 and Subject 3.
- **Find the roll number of the student with maximum marks in Subject 3**: Identify the student with the highest marks in Subject 3 and print their roll number.
- **Find the roll number of the student with minimum marks in Subject 2**: Identify the student with the lowest marks in Subject 2 and print their roll number.
- **Find the roll number of students who scored 24 marks in Subject 2**: Identify students who obtained exactly 24 marks in Subject 2 and print their roll numbers.
- **Find the count of students who got less than 40 marks in Subject 1**: Count the number of students who scored less than 40 marks in Subject 1.
- **Find the count of students who got more than 90 marks in Subject 2**: Count the number of students who scored more than 90 marks in Subject 2.
- **Find the count of students who scored >=90 in each subject**: Count the number of students who scored 90 or more marks in each subject.
- **Find the count of subjects in which each student scored >=90**: Determine how many subjects each student scored 90 or more marks in.
- **Print Subject 1 marks in ascending order**: Sort and print the marks of students in Subject 1 in ascending order.
- **Print students who scored between 50 and 90 in Subject 1**: Display students who scored marks between 50 and 90 in Subject 1.

- **Find index positions of students who scored 79 in Subject 1**: Identify the index positions of students who scored exactly 79 marks in Subject 1.

CODE:

```
import numpy as npa = np.loadtxt("Sample.csv", delimiter=',',
skiprows=1)

# 1. Print all student details

print("All student Details:\n",a)

# 2. print total students

print("Total Students:",a.shape[0]) # 3. Print

all student Roll numbers print("All Student

Roll Nos",a[:,0]) # 4. Print subject 1 marks

print("Subject 1 Marks",a[:,1]) # 5. print

minimum marks of Subject 2 print("Min

marks in Subject 2",np.min(a[:,2])) # 6.

print maximum marks of Subject 3

print("Max marks in Subject

3",np.max(a[:,3]))

# 7. Print All subject marks print("All

subject marks:",a[:,1:4]) # 8. print Total

marks of students print("Total

Marks",np.sum(a[:,1:4],axis=1))

# 9. print average marks of each student

#Note:Expected output shows rounded values like 77.3

print(np.round(np.mean(a[:,1:4],axis=1),1)) # 10. print
```

average marks of each subject print("Average Marks

of each

subject",np.round(np.mean(a[:,1:4],axis=0),1))

11. print average marks of S1 and S2

#The test case expected[71.7 59.8], which are the individual subject averages

print("Average Marks of S1 and S2",np.round(np.mean(a[:,1:3],axis=0),1)) # 12. print

average marks of S1 and S3 print("Average Marks of S1

and S3",np.round(np.mean(a[:,[1, 3]],axis=0),1))

13. print Roll number who got maximum marks in Subject 3

print("Roll no who got maximum marks in Subject

3",a[np.argmax(a[:,3]),0])

14. print Roll number who got minimum marks in Subject 2

print("Roll no who got minimum marks in Subject

2",a[np.argmin(a[:,2]),0])

15. print Roll number who got 24 marks in Subject 2

print("Roll no who got 24 marks in Subject 2",a[a[:,2]==24]

[:,0:1]) # 16. print count of students who got marks in

Subject 1 < 40 print("Count of students who got marks in

Subject 1 < 40",np.sum(a[:,1]<40))

17. print count of students who got marks in Subject 2 > 90

print("Count of students who got marks in Subject 2 >

90:",np.sum(a[:,2]>90))

18. print count of students in each subject who got marks

>= 90 print("Count of students in each subject who got marks

>= 90:",np.sum(a[:,1:4]>=90,axis=0))# 19. print count of subjects in which each student got marks >= 90 print("Roll no:",a[:,0])

print("Count of subjects in which student got marks >= 90:",np.sum(a[:,1:4]>=90,axis=1)) # 20. Print S1 marks in ascending order print(np.sort(a[:,1]))

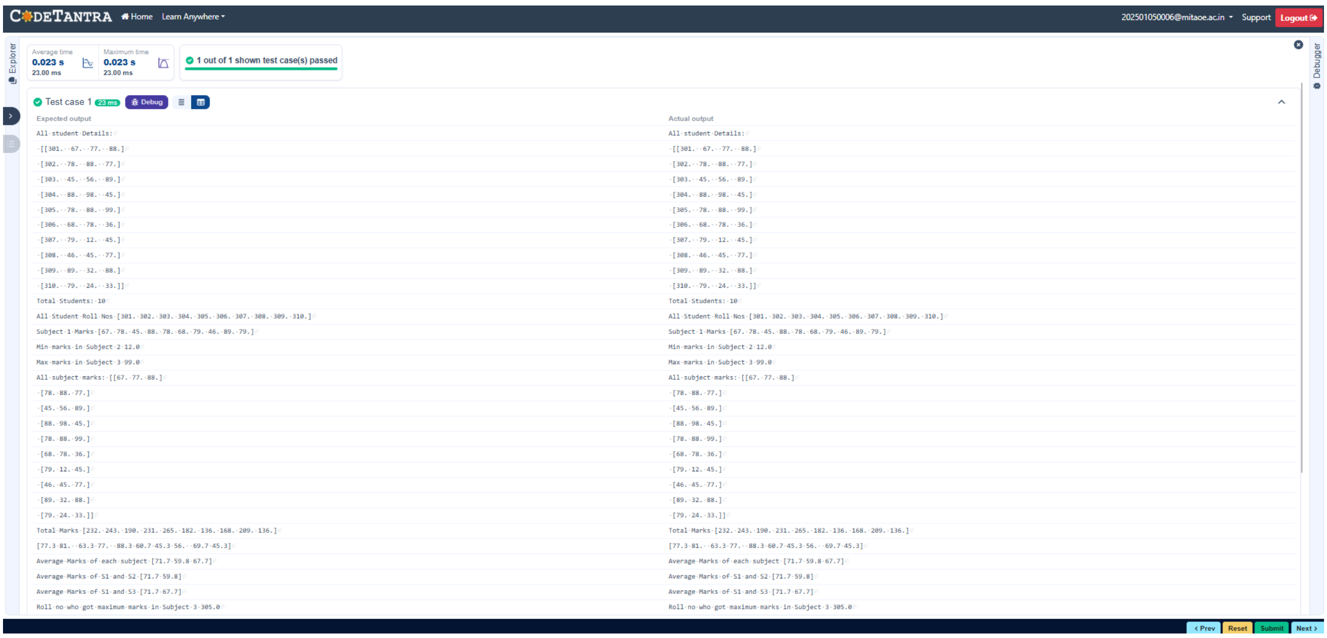
21. Print S1 marks >= 50 and <= 90print(a[(a[:,1]>=50)&(a[:,1]<=90)])

#Bonus: The test case shows the full array printed again before the index positions print(a)

22. Print the index position of marks 79

Expected format:(array([6,9],dtype=int32),)

print(np.where(a[:,1]==79))



Explorer

Average time
0.014 s
14.00 ms

Maximum time
0.014 s
14.00 ms

1 out of 1 shown test case(s) passed

```
 [78. 88. 99.]
 [68. 78. 36.]
 [79. 12. 45.]
 [46. 45. 77.]
 [89. 32. 88.]
 [79. 24. 33.]]
Total Marks [232. 243. 190. 231. 265. 182. 136. 168. 209. 136.]
[77.3 81.  63.3 77.  88.3 60.7 45.3 56.  69.7 45.3]
Average Marks of each subject [71.7 59.8 67.7]
Average Marks of S1 and S2 [71.7 59.8]
Average Marks of S1 and S3 [71.7 67.7]
Roll no who got maximum marks in Subject 3 305.0
Roll no who got minimum marks in Subject 2 307.0
Roll no who got 24 marks in Subject 2 [[310.]]
Count of students who got marks in Subject 1 < 40 0
Count of students who got marks in Subject 2 > 90: 1
Count of students in each subject who got marks >= 90: [0 1 1]
Roll no: [301. 302. 303. 304. 305. 306. 307. 308. 309. 310.]
Count of subjects in which student got marks >= 90: [0 0 0 1 1 0 0 0 0 0]
[45. 46. 67. 68. 78. 78. 79. 79. 88. 89.]
[[301.  67.  77.  88.]
 [302.  78.  88.  77.]
 [304.  88.  98.  45.]
```

```
 [78. 88. 99.]
 [68. 78. 36.]
 [79. 12. 45.]
 [46. 45. 77.]
 [89. 32. 88.]
 [79. 24. 33.]]
Total Marks [232. 243. 190. 231. 265. 182. 136. 168. 209. 136.]
[77.3 81.  63.3 77.  88.3 60.7 45.3 56.  69.7 45.3]
Average Marks of each subject [71.7 59.8 67.7]
Average Marks of S1 and S2 [71.7 59.8]
Average Marks of S1 and S3 [71.7 67.7]
Roll no who got maximum marks in Subject 3 305.0
Roll no who got minimum marks in Subject 2 307.0
Roll no who got 24 marks in Subject 2 [[310.]]
Count of students who got marks in Subject 1 < 40 0
Count of students who got marks in Subject 2 > 90: 1
Count of students in each subject who got marks >= 90: [0 1 1]
Roll no: [301. 302. 303. 304. 305. 306. 307. 308. 309. 310.]
Count of subjects in which student got marks >= 90: [0 0 0 1 1 0 0 0 0 0]
[45. 46. 67. 68. 78. 78. 79. 79. 88. 89.]
[[301.  67.  77.  88.]
 [302.  78.  88.  77.]
 [304.  88.  98.  45.]
```

Debugger

‹ Prev Reset Submit Next ›

Explorer

Average time
0.014 s
14.00 ms

Maximum time
0.014 s
14.00 ms

1 out of 1 shown test case(s) passed

```
Roll no: [301. 302. 303. 304. 305. 306. 307. 308. 309. 310.]
Count of subjects in which student got marks >= 90: [0 0 0 1 1 0 0 0 0 0]
[45. 46. 67. 68. 78. 78. 79. 79. 88. 89.]
[[301.  67.  77.  88.]
 [302.  78.  88.  77.]
 [304.  88.  98.  45.]
 [305.  78.  88.  99.]
 [306.  68.  78.  36.]
 [307.  79.  12.  45.]
 [309.  89.  32.  88.]
 [310.  79.  24.  33.]]
[[301.  67.  77.  88.]
 [302.  78.  88.  77.]
 [303.  45.  56.  89.]
 [304.  88.  98.  45.]
 [305.  78.  88.  99.]
 [306.  68.  78.  36.]
 [307.  79.  12.  45.]
 [308.  46.  45.  77.]
 [309.  89.  32.  88.]
 [310.  79.  24.  33.]]
(array([6, 9], dtype=int32),)
```

```
Roll no: [301. 302. 303. 304. 305. 306. 307. 308. 309. 310.]
Count of subjects in which student got marks >= 90: [0 0 0 1 1 0 0 0 0 0]
[45. 46. 67. 68. 78. 78. 79. 79. 88. 89.]
[[301.  67.  77.  88.]
 [302.  78.  88.  77.]
 [304.  88.  98.  45.]
 [305.  78.  88.  99.]
 [306.  68.  78.  36.]
 [307.  79.  12.  45.]
 [309.  89.  32.  88.]
 [310.  79.  24.  33.]]
[[301.  67.  77.  88.]
 [302.  78.  88.  77.]
 [303.  45.  56.  89.]
 [304.  88.  98.  45.]
 [305.  78.  88.  99.]
 [306.  68.  78.  36.]
 [307.  79.  12.  45.]
 [308.  46.  45.  77.]
 [309.  89.  32.  88.]
 [310.  79.  24.  33.]]
(array([6, 9], dtype=int32),)
```

Debugger

‹ Prev Reset Submit Next ›